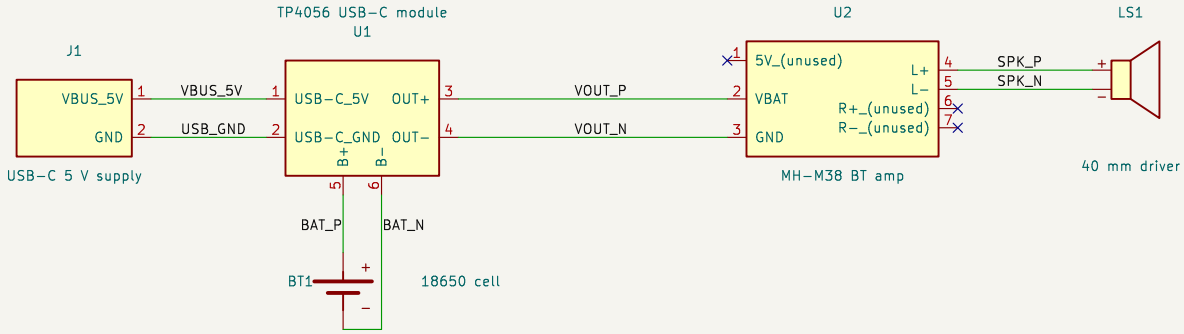


NOTES
1. Cell connects to the module's B+/B- only. The load (amp VBAT/GND) connects to OUT+/OUT- so the protection IC is in the load path.
2. One speaker on ONE channel (L+/L-). Leave R+/R- open and never tie L- and R- together.
3. Do not use the amp's micro-USB or its 5V pad while the battery is on VBAT.
4. DESIGN GAP: no power switch. The amp powers up whenever the cell is connected. A slide switch in series with
   VOUT_P -> VBAT would add an off state (needs an enclosure cutout).
5. The TP4056 module does not do true load-sharing: charge with the speaker off to get a proper full-charge indication.
6. Before connecting the cell, confirm the charge module has THREE chips (TP4056, DW01A, 8205A).
7. Connect the cell last and check polarity with a meter.

PARTS
J1 any 5 V USB-C supply, >= 1 A
U1 TP4056 Type-C module with DW01A + 8205A protection, ~28 x 17 mm (ShillehTek or equivalent)
U2 MH-M38 Bluetooth 4.2 amp, 45 x 20 mm (DFRobot DFR0720 is this module)
BT1 Samsung INR18650-25R, flat top, UNPROTECTED
LS1 40 mm 4 ohm driver, 20 mm tall or less (Adafruit 3968 or equivalent)

Status: design only, not built or tested
Module-level wiring; no custom PCB

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Title: 18650 Bluetooth Speaker - wiring schematic

Size: A4	Date: 2026-10-26	**Rev: A**
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